

Short communication

# Poultry abattoir survey of carcass condemnations in Fars province, southern Iran

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## Abstract

Official post-mortem inspection records of poultry in 11 industrial poultry abattoirs in the Fars province, southern Iran, between 20 March 2002 and 19 March 2006 were used to investigate the character and prevalence of poultry loss caused by disease and pathological changes in the province. Linear regression analysis for study of time trend and  $\chi^2$ -test for investigation of seasonal pattern were used as statistical methods. In the study period, 130,967,021 birds were slaughtered. As a result of official veterinary inspection, 959,416 (0.73%) birds were condemned. The condemnation risk was calculated for each quarter in a year. Cachexia and septicemia were the most common reasons for rejection of carcasses and were responsible for 62% of the total condemnations. The condemnation risk due to septicemia increased over the study period from 0.14 to 0.22%. The rejection risk due to overscalding declined over the study period from 0.07 to 0.01%. No trends in time were observed for the other causes of condemnation. In conclusion the majority of condemnations are caused by disease and improving the standard of disease prevention and control on farms is necessary. These results could form the basis of an assessment of trends in condemnations in poultry slaughterhouse and should prove useful to the poultry industry as a baseline data for future comparison.

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## 1. Introduction

Poultry meat production throughout the world continues to expand both in amount and in sophistication. In Iran as all over the world the trend in poultry production has been increasing over the last two decades. At present, annual poultry meat production is more than one million tonnes in the country and yet there is still further capacity for production. Among the 28 provinces, Fars province in southern Iran has been rated fifth largest producer of poultry meat in the country and is responsible for 6% of total poultry meat production. In this province as other parts of the country, commercial chicken–meat flocks are from industrial breeds like Ross, Cobb and Arbor Acor. Feed starter, grower and finisher diets are primarily based on corn and soybeans, which may include growth promotants, antibiotics, and anticoccidials. They are slaughtered at 45–55 days of age.

In the poultry production industry, disease and pathological changes represent a serious problem. Apart from economic losses, disease in poultry might constitute an epidemiologic and zoonotic threat and problems concerning meat hygiene and possible health risk to the consumer should be deliberated. In this context, meat-inspection data are a potential source of information and have an important role to play in epidemiology and preventive veterinary medicine (Gracey et al., 1999). Monitoring disease and other conditions at slaughter has been recognized as one way of assessing the disease status of a herd or flock (Herenda and Jakel, 1994), however this source of information is not being fully exploited worldwide. The intent of the present study was to investigate the character and prevalence of poultry loss caused by disease and pathological changes in Fars province, southern Iran, and to analyze the dynamics of trends in the prevalence of poultry carcass condemnation during a 4-year period. The results should prove useful to the poultry industry as a baseline data for future comparison.

## 2. Material and methods

Official post-mortem inspection records of poultry in 11 industrial poultry abattoirs in the Fars province, southern Iran, during 20 March 2002–19 March 2006 (equal to 1381–1384 H.S., Iranian calendar) were obtained from the veterinary services of the province. These records were used as the sources of data from which the condemnation data were extracted. Data consisted of the total number slaughtered, total number condemned and number of those condemned for specific conditions on a monthly basis. Specific disease conditions consisted of ascites, bronchitis/CRD, bruising, cachexia, Marek's disease, poisoning, septicemia, synovitis/arthritis and tuberculosis. Moreover, slaughter-related conditions consisted of contamination and overscalding. Poultry carcasses may be contaminated in various ways during the slaughter process, such as fecal matters or dirty scalding tank water. Overscalding refers to carcasses which present a cooked appearance due to being held too long in the scalding tank. Decision for rejection were made by trained poultry meat inspectors on the basis of visual inspection of pathologic changes.

Linear regression analysis was performed to study the time trend. One regression analysis was conducted for each cause of condemnation. Here the risk of condemnation

evaluated for a 3-month period (quarter a year) was used as the response. Seasonal pattern over the full study period was investigated by  $\chi^2$ -test. Differences in condemnation risks for disease conditions and for slaughter-related causes of condemnation were compared between seasons. To reduce the risk of type I error in such a large data set, a  $P$ -value less than 0.01 was considered statistically significant.

### 3. Results

In the study period, almost 131 million birds were slaughtered in the province. As a result of inspections, 959,416 (0.73%) birds were condemned. Cachexia and septicemia were the most common reasons for rejection of carcasses, respectively. They were responsible for 0.62% of the total condemnations in this period (Table 1). No tuberculosis cases were seen and synovitis/arthritis and Marek's disease were uncommon reasons for condemnation.

According to the results of the regression analysis, significant increase was observed in condemnation risk due to septicemia over the study period from 0.14 to 0.22% ( $R^2 = 0.70$ ,  $P < 0.001$ ), on the other hand, rejection risk due to overscalding declined significantly from 0.07 to 0.01% ( $R^2 = 0.79$ ,  $P < 0.001$ ) (Fig. 1). No significant trends in time were observed for the total condemnation risk or other causes of condemnation during the study period (Fig. 2a and b).

Condemnation for disease conditions had a significant seasonal pattern, with highest risks occurring in the colder months (Table 2a). Condemnation due to slaughter-related causes was most common during April–June (Table 2b).

Table 1  
Causes of carcass condemnations in industrial poultry slaughterhouses in the Fars province, southern Iran, during 2002–2006

| Cause of condemnation | Poultry condemned |       | Relative % of condemnations |
|-----------------------|-------------------|-------|-----------------------------|
|                       | No.               | %     |                             |
| Cachexia              | 361,973           | 0.280 | 37.7                        |
| Septicemia            | 233,314           | 0.180 | 24.3                        |
| Poisoning             | 128,151           | 0.098 | 13.3                        |
| Bronchitis/CRD        | 115,852           | 0.088 | 12.1                        |
| Bruises               | 17,442            | 0.013 | 1.8                         |
| Ascites               | 15,118            | 0.012 | 1.6                         |
| Marek's disease       | 7,283             | 0.006 | 0.7                         |
| Synovitis/arthritis   | 2,548             | 0.002 | 0.3                         |
| Tuberculosis          | 0                 | 0.000 | 0.0                         |
| Overscalding          | 52,318            | 0.040 | 5.5                         |
| Contamination         | 20,342            | 0.016 | 2.1                         |
| Miscellaneous         | 5,075             | 0.004 | 0.5                         |
| Total condemnations   | 959,416           | 0.732 | 100                         |
| Total slaughtered     | 130,967,021       |       |                             |

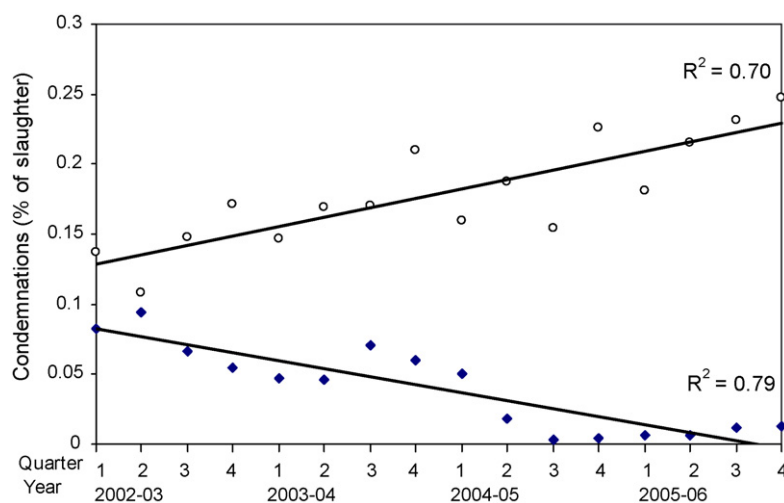


Fig. 1. Trends in frequency of septicemia (open circles) and overscalding (closed circles) as a reason for condemnation of poultry in the Fars province, southern Iran, during 2002–2006 based on the slaughter of 131 million birds.

#### 4. Discussion

Diet, breed and management of poultry may be associated with the presence or absence of pathological lesions and could cause carcass condemnation during the inspection process. It was shown that the proportion of birds totally condemned and reasons for condemnations differ between countries (Radkowski et al., 1996). This is reasonable due to different ecologic condition, epidemiologic aspects and management practice in each country.

In the present study, the average condemnation risk was 0.73%. This corresponded relatively well with findings in the literature. A survey of the prevalence and type of poultry carcass condemnation was carried out in Poland during 1986–1991 (Radkowski et al., 1996). Around 38 m carcasses examined, 1.66% was condemned and the most common reason for rejection was Marek's disease. In England and Wales during April 1992–March 1993, 1.3% of slaughtered broilers were condemned and septicemia/toxaemia/fever were the most frequent causes of condemnation (Bremner, 1994).

Table 2a

Seasonal slaughter and condemnation for disease conditions in industrial poultry slaughterhouses in the Fars province, southern Iran, during 2002–2006

|                  | Number slaughtered | Number condemned (%) |
|------------------|--------------------|----------------------|
| April–June       | 29811529           | 177206 (0.59)        |
| July–September   | 29872893           | 184471 (0.62)        |
| October–December | 34557867           | 231513 (0.67)        |
| January–March    | 36724732           | 288491 (0.78)        |
| Total            | 130967021          | 881681 (0.67)        |

$\chi^2 = 11089$ ,  $df = 3$ ,  $P < 0.001$ .

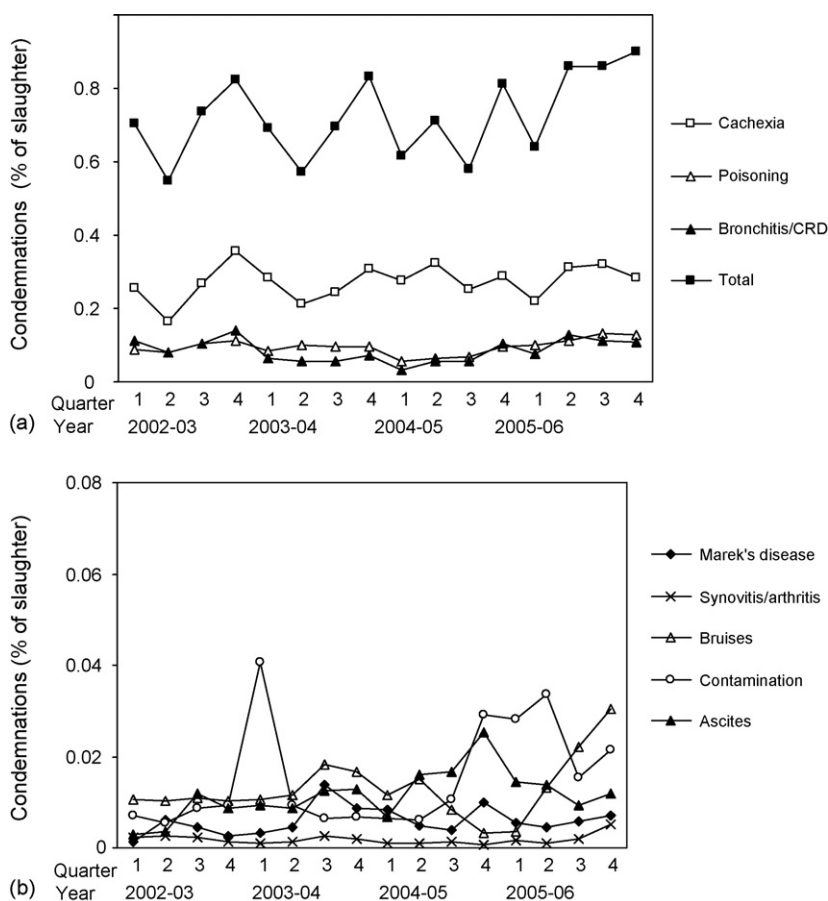


Fig. 2. (a and b) Trends in frequency of various causes of condemnation of poultry in the Fars province, southern Iran, between 2002 and 2006 based on the slaughter of 131 million birds.

Table 2b

Seasonal slaughter and condemnation for slaughter-related causes of condemnation in industrial poultry slaughterhouses in the Fars province, southern Iran, during 2002–2006

|                  | Number slaughtered | Number condemned (%) |
|------------------|--------------------|----------------------|
| April–June       | 29811529           | 20623 (0.07)         |
| July–September   | 29872893           | 16650 (0.05)         |
| October–December | 34557867           | 17013 (0.05)         |
| January–March    | 36724732           | 18374 (0.05)         |
| Total            | 130967021          | 72660 (0.05)         |

$\chi^2 = 1449$ ,  $df = 3$ ,  $P < 0.001$ .

In the present study the most frequent causes of condemnation were cachexia and septicemia. In industrial poultry production chickens are fed ad libitum, therefore cachexia is primarily secondary to disease and pathologic conditions, and it is unlikely to be a result of malnutrition. A significant increasing trend during the study period was observed for septicemia (Fig. 1). Septicemia is a rather non-specific term, since it includes birds that were rejected because of congested, darkened muscles with inflammatory lesions such as airsacculitis and perihepatitis that may be an indication of disease, but it does not specify what the disease might be (Gracey et al., 1999; Austic and Nesheim, 1990). Among microbial causes of septicemia *Escherichia coli*, *Salmonella enteritidis* and *Pasteurella multocida* are important pathogens of human relevance. Therefore, as suggested by Fisher et al. (1998), early identification of septicemia might have important public health implications. In the present work, unfortunately no bacteriologic examination on septicemic carcasses was done. Because the organisms present tend to be common to the whole flock (Gracey et al., 1999), identifying the microorganisms causing disease is highly recommended.

Overscalding and contamination are related to the slaughter process and therefore expected to be under control in the slaughterhouses. With this regard, while overscalding declined during the study period, lack of change in the contamination risk was disappointing and indicates that more work needs to be done to prevent contamination of carcasses at these abattoirs.

The high level of condemnation due to diseases in colder months is logical. More rejection of contaminated and overscald carcasses in spring would be attributed to coincidence with the national holidays time in the country. In this time, fast slaughter rate and probable decrease in work force may be responsible factors.

In conclusion, the majority of condemnations are caused by disease. The control and prevention of disease, however, has to be on the farms and in the hatcheries. Therefore, improving the standard of disease prevention and control on farms is necessary. These results could form the basis of an assessment of trends in condemnations in poultry slaughterhouses in the future.

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